

DRISHTI

Government Deployment & Operational Adoption Blueprint

From Proof-of-Concept to Gujarat-Scale Intelligent CCTV Infrastructure

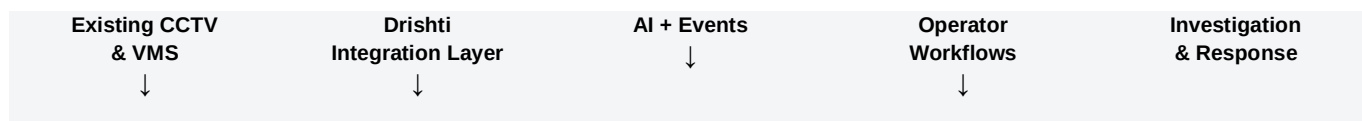
Purpose: A practical blueprint for how Drishti could be piloted, governed, operated, evaluated, and progressively expanded across existing government CCTV infrastructure.

Sentinel Gujarat • Additional Submission Document

This document complements the Solution Presentation and High-Level Design; it focuses on adoption and operations rather than repeating the architecture.

1. Executive Deployment Vision

Drishti is positioned as an intelligence and orchestration layer over existing CCTV infrastructure. A government deployment should therefore avoid a “replace everything first” approach. Instead, cameras and departments can be onboarded progressively, with the platform proving interoperability, operational value, AI usefulness, security, and reliability at each stage.



The deployment principle is simple: preserve useful existing infrastructure, standardize access through Drishti, turn video into searchable events, and expand only after measurable pilot outcomes are demonstrated.

Deployment principles

- Integrate before replacing: use existing cameras, feeds, VMSs and networks wherever technically feasible.
- Pilot with diversity: include different vendors, codecs, resolutions, locations and network conditions.
- Human-in-the-loop: AI should prioritize and organize information; authorized personnel remain responsible for operational decisions.
- Evidence before scale: each phase should produce technical and operational evidence for the next phase.
- Separate current capability from future production architecture: PoC behavior must not be presented as statewide capacity.

2. Adoption Journey: PoC → Pilot → Statewide

A staged rollout reduces technical and organizational risk. The same platform can mature while the number of cameras and participating departments increases.

PoC	Controlled Pilot	City / Department Expansion	Regional Deployment	Statewide Platform
↓	↓	↓	↓	
Stage	Primary objective	Typical activities	Go / No-Go evidence	
PoC	Prove the core workflow	Feed ingestion, AI detection, ANPR, alerts, GIS, audit	Core workflow is repeatable	
Pilot	Prove interoperability and operations	Diverse cameras, operators, watchlists, failure handling	Stable operation under representative conditions	
City / Department	Prove multi-department value	More cameras, department ownership, role-based access	Operational procedures work at larger scope	
Regional	Prove distributed processing	Regional GPU/edge architecture, metadata transport, HA	Regional scale and recovery behavior validated	
Statewide	Institutionalize the platform	Governance, production security, statewide onboarding	Production readiness and governance sign-off	

3. Recommended Pilot Design

The first government pilot should not simply choose the easiest cameras. It should deliberately test the conditions that the statewide system will eventually face.

Pilot dimension	What to include	Why it matters
Camera diversity	Multiple vendors / VMS types and protocols	Tests interoperability rather than one-vendor success
Locations	Traffic junctions, public areas, offices/checkpoints where permitted	Tests different scene types
Image quality	Good and difficult feeds	Tests confidence handling and operator verification
Network	Stable and constrained links	Tests reconnect and bandwidth behavior
AI workload	Vehicle-heavy and mixed scenes	Tests detection, tracking and ANPR workflow
Operations	Different authorized user roles	Tests RBAC and practical workflows

Failure cases	Camera/network/service interruptions	Tests recovery rather than only happy-path behavior
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Pilot success should be measured, not assumed

The pilot should establish a baseline and then measure:

- camera onboarding success rate
- stream availability and reconnect behavior
- useful ANPR read rate by camera condition
- alert latency and false-alert rate
- operator workload and investigation workflow time
- GPU utilization and processing load
- network/bandwidth behavior
- system availability and recovery

4. 30 / 60 / 90-Day Implementation Roadmap

Period	Focus	Key outputs
Days 0–30	Discovery & integration	Camera inventory; connectivity assessment; protocol/authentication mapping; pilot onboarding; baseline measurements
Days 31–60	Operational pilot	AI policies; ANPR/watchlist workflow; alerts; GIS investigation workflow; RBAC; operator training; failure drills
Days 61–90	Production-readiness assessment	Scale test; reliability results; security review; operating procedures; lessons learned; expansion decision

Discover



Integrate



Operate



Measure



Harden



Expand

5. Department & Camera Onboarding Playbook

Every department should follow the same onboarding lifecycle. This turns heterogeneous CCTV infrastructure into a predictable operational process.

Department Request	Camera Inventory	Protocol / VMS Discovery	Connectivity + Authentication	Catalogue Registration	Health Check	AI Policy + Operator Access	Go Live
↓	↓	↓	↓	↓	↓	↓	

Camera onboarding record

Field group	Examples / purpose
Identity	Camera ID, department, ownership, site
Location	GPS coordinates, address/zone, operational area
Video	RTSP/HLS/WebRTC availability, codec, resolution, source characteristics
Connectivity	Network path, latency/reliability considerations, firewall/NAT requirements
Security	Approved authentication method; credentials remain server-side
AI policy	Enabled analytics, sampling policy, watchlist applicability
Operations	Health status, maintenance owner, escalation path

6. AI Governance: What Happens When AI Is Wrong?

A government surveillance platform must treat AI output as decision support, not as an unquestionable conclusion. Low-confidence detections and OCR should remain identifiable as uncertain, and operational actions should be subject to authorized human verification.

Video	AI Detection	Confidence / Quality Check	Event / Alert	Human Verification	Confirm / Dismiss	Audit
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Situation	System behavior	Operator responsibility
High-confidence useful event	Surface event prominently	Verify context and take authorized action
Uncertain plate OCR	Retain uncertainty / avoid forced plate identity	Review frame or additional sightings
Watchlist match	Generate alert according to configured policy	Validate match before operational escalation
Repeated sighting	Associate with existing vehicle track/event where supported	Use journey context rather than a single frame
False alert	Allow dismissal / resolution and record action	Provide feedback and preserve audit trail

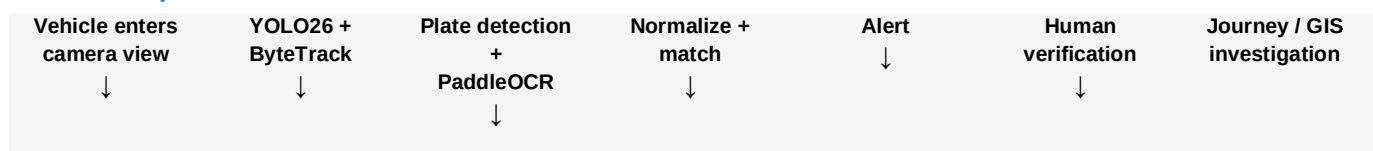
7. Data Lifecycle & Evidence Discipline

Drishti should distinguish transient video processing from persistent intelligence. Not every frame needs to become a stored record. The operational value comes from retaining the right metadata and selected evidence while respecting the applicable retention and access policy.

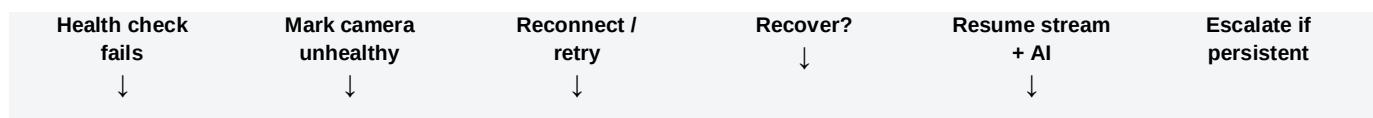
Live Camera	Frame / Stream Processing	Detection / ANPR Event	Investigation Record	Selected Evidence	Retention / Archive / Deletion Policy
↓	↓	↓	↓	↓	
Data type	Role	Recommended treatment			
Live video	Real-time observation and AI input	Process as required; do not imply automatic long-term storage			
AI metadata	Searchable intelligence: camera, timestamp, track, plate where reliable	Store according to operational and policy requirements			
Alerts	Prioritized operational events	Persist status, resolution and audit information			
Investigation records	Human-led case context	Access-controlled and retained according to applicable policy			
Evidence	Selected material needed for an investigation	Controlled evidence storage with integrity/access controls			

8. Operational Playbooks

8.1 Stolen / Watchlisted Vehicle

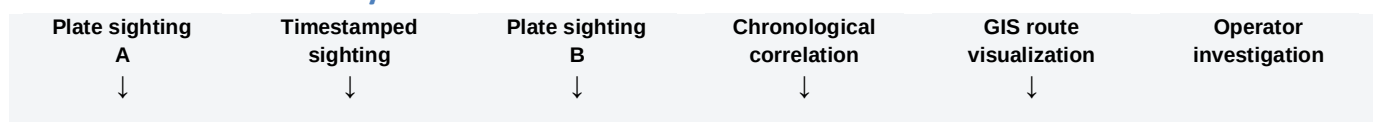


8.2 Camera Goes Offline



The important operational principle is that camera failure should become an observable system event, not a silent gap.

8.3 Multi-Camera Journey



9. Human + AI Operating Model

Role	Primary responsibility	Drishti support
Control-room operator	Monitor prioritized events and live cameras	Camera wall, alerts, verification
Traffic officer / command center	Investigate traffic-related sightings	ANPR, journey search, GIS
Crime branch / cyber cell	Investigate vehicle movement and watchlist events	Historical search, correlation, audit trail
System administrator	Manage users, permissions, system configuration	RBAC, health and administration
Department camera owner	Maintain source-camera information and operational status	Catalogue and ownership metadata

This division prevents the platform from becoming “AI-only.” Drishti is most useful when it connects machine-generated events with a clear human operating procedure.

10. Scaling the Operating Model

The approximate 80,000-camera challenge scale should be treated as a production design target, not as a claim about the capacity of the current PoC workstation. The adoption model should therefore scale the operating architecture in stages.

Camera / NVR Edge	Regional Processing	Central Intelligence	Department / Operator	Investigation / Response
↓	↓	↓	↓	
Layer	Primary role	Why it scales		
Edge / site	Ingest, basic processing and local resilience where appropriate	Avoids sending every raw frame to the center		
Regional	GPU-heavy inference, track/metadata processing	Pools compute and reduces central bottlenecks		
Central	Search, watchlists, correlation, GIS, governance and operator services	Centralizes intelligence rather than all raw video		
Operator	Consumes prioritized events instead of continuously inspecting every feed	Reduces manual monitoring burden		

11. Security & Governance Checklist

Control area	Deployment expectation
Identity	Role-based access with named users and appropriate administrative controls
Streaming credentials	Keep upstream credentials server-side; never expose them in browser URLs or logs
Transport security	Use secure transport and service-to-service protections in production
Audit	Record security-sensitive and operational actions in an auditable trail
Least privilege	Give departments/operators only the camera/data access required for their role
AI governance	Record confidence/uncertainty and preserve human verification for consequential actions
Secrets	Use protected environment/secret management; never commit credentials
Production hardening	Apply SSO/AD, mTLS, HSM and stronger key management where required by deployment policy

12. Risk Register

Risk	Impact	Mitigation / design response
Legacy / heterogeneous cameras	High	Protocol/VMS integration layer; phased onboarding
Poor image quality	High	Quality checks, confidence handling, human verification
Unstable connectivity	High	Reconnect strategy, health monitoring, edge/regional processing
AI false positives	High	Thresholds, contextual correlation, human-in-loop, audit
Vendor lock-in	Medium/High	Hardware-agnostic integration and standardized metadata
Central compute bottleneck	High	Edge/regional GPU architecture and distributed processing
Credential leakage	High	Server-side proxying and secret management
Database / service failure	High	HA, backups, monitoring and recovery procedures in production
Operator overload	Medium/High	Event prioritization, filters, watchlists and workflow design

13. Go-Live Checklist

- Camera ownership and department contact are recorded.
- Source connectivity and authentication have been validated.
- Camera metadata and location are complete enough for operations.
- Live viewing works through the approved secure path.
- AI processing policy is explicitly configured for the camera.
- ANPR behavior and uncertainty handling are understood by operators.
- Watchlist source and update process are defined.
- Alert escalation and resolution workflow is defined.
- User roles and least-privilege access are configured.
- Audit logging is enabled and review responsibility is assigned.
- Failure/reconnect behavior has been exercised.
- Retention/evidence rules are approved before production use.
- Pilot results are recorded before adding the next deployment cohort.

14. What Gujarat Gets From This Approach

The main benefit is not simply a new camera viewer. The deployment model creates a common operational layer over fragmented CCTV infrastructure.

Existing challenge	Deployment outcome with Drishti
Fragmented camera systems	Common catalogue, access and operational workflow
Manual monitoring	Prioritized events and searchable vehicle intelligence
Manual vehicle investigation	ANPR history and multi-camera journey reconstruction where data permits
Different departments	Department-aware ownership and access model
Difficult statewide scaling	Edge/regional/central architecture that can grow incrementally
Unclear AI accountability	Human verification, confidence handling and audit trail
High migration risk	Phased onboarding instead of immediate infrastructure replacement

15. Recommended Next-Step Sequence



Each expansion cohort should carry forward the lessons, integration templates, operating procedures, security controls and measured results from the previous stage. This turns statewide deployment into a repeatable engineering process rather than a single high-risk migration.

16. Final Position

Drishti should be adopted as a progressive intelligence layer over Gujarat’s existing CCTV ecosystem: integrate first, prove value, measure operational behavior, harden the platform, and then scale.

The approach deliberately separates what can be demonstrated in the current PoC from what belongs in a production statewide architecture. That distinction makes the deployment plan more credible: the pilot proves the workflow, while the production roadmap provides a controlled path toward distributed processing, stronger security, multi-department governance and large-scale operations.

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